Name: ____

A Level Mathematics : Statistics

Topic 3: Probability

Date:

Time:

Total marks available:

Total marks achieved: ______

Questions

Q1.

Magali is studying the mean total cloud cover, in oktas, for Leuchars in 1987 using data from the large data set. The daily mean total cloud cover for all 184 days from the large data set is summarised in the table below.

Daily mean total cloud cover (oktas)	0	1	2	3	4	5	6	7	8
Frequency (number of days)	0	1	4	7	10	30	52	52	28

One of the 184 days is selected at random.

(a) Find the probability that it has a daily mean total cloud cover of 6 or greater.

(1)

Magali is investigating whether the daily mean total cloud cover can be modelled using a binomial distribution.

She uses the random variable X to denote the daily mean total cloud cover and believes that $X \sim B(8, 0.76)$

Using Magali's model,

(b) (i) find $P(X \geq 6)$

(2)

(ii) find, to 1 decimal place, the expected number of days in a sample of 184 days

with a daily mean total cloud cover of 7

(2)

(c) Explain whether or not your answers to part (b) support the use of Magali's model.

(1)

There were 28 days that had a daily mean total cloud cover of 8
For these 28 days the daily mean total cloud cover for the **following** day is shown in the table below.

Daily mean total cloud cover (oktas)	0	1	2	3	4	5	6	7	8
Frequency (number of days)	0	0	1	1	2	1	5	9	9

(d) Find the proportion of these days when the daily mean total cloud cover was 6 or greater.

(1)

(e) Comment on Magali's model in light of your answer to part (d).

(2)

(Total for question = 9 marks)

Q2.

Three bags, *A*, *B* and *C*, each contain 1 red marble and some green marbles.

Bag *A* contains 1 red marble and 9 green marbles only

Bag *B* contains 1 red marble and 4 green marbles only

Bag *C* contains 1 red marble and 2 green marbles only

Sasha selects at random one marble from bag *A*.
If he selects a red marble, he stops selecting.
If the marble is green, he continues by selecting at random one marble from bag *B*.
If he selects a red marble, he stops selecting.
If the marble is green, he continues by selecting at random one marble from bag *C*.

(a) Draw a tree diagram to represent this information.

(2)

(b) Find the probability that Sasha selects 3 green marbles.

(2)

(c) Find the probability that Sasha selects at least 1 marble of each colour.

(2)

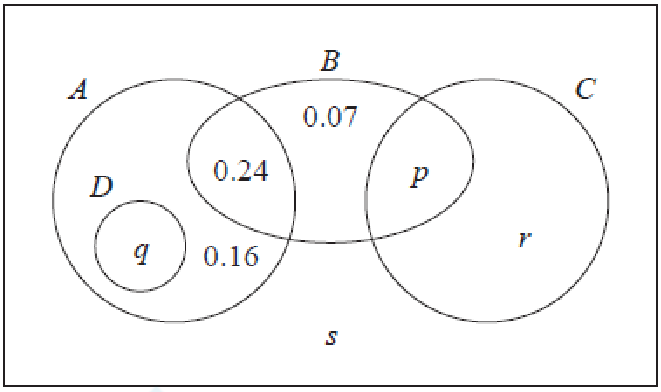
(d) Given that Sasha selects a red marble, find the probability that he selects it from bag *B*.

(2)

(Total for question = 8 marks)

Q3.

The Venn diagram shows the probabilities associated with four events, *A*, *B*, *C* and *D*



(a) Write down any pair of mutually exclusive events from *A*, *B*, *C* and *D*

(1)

Given that P(*B*) = 0.4

(b) find the value of *p*

(1)

Given also that *A* and *B* are independent

(c) find the value of *q*

(2)

Given further that P(*B*' | *C*) = 0.64

(d) find

(i) the value of *r*

(ii) the value of *s*

(4)

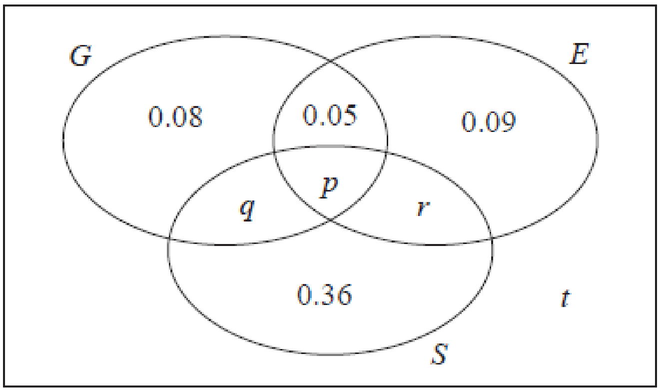
(Total for question = 8 marks)

Q4.

A large college produces three magazines.
One magazine is about green issues, one is about equality and one is about sports.
A student at the college is selected at random and the events *G*, *E* and *S* are defined as follows

G is the event that the student reads the magazine about green issues
E is the event that the student reads the magazine about equality
S is the event that the student reads the magazine about sports

The Venn diagram, where p, q, r and t are probabilities, gives the probability for each subset.



(a) Find the proportion of students in the college who read exactly one of these magazines.

(1)

No students read all three magazines and P(G) = 0.25

(b) Find

(i) the value of p

(ii) the value of q

(3)

Given that $P(S \mid E) = \frac{5}{12}$

(c) find

(i) the value of r

(ii) the value of t

(4)

(d) Determine whether or not the events ($S \cap E'$) and G are independent.

Show your working clearly.

(3)

(Total for question = 11 marks)

Q5.

Given that

$$P(A) = 0.35 \qquad P(B) = 0.45 \qquad \text{and} \qquad P(A \cap B) = 0.13$$

find

(a) P(*A*′ | *B*′)

(2)

(b) Explain why the events *A* and *B* are not independent.

(1)

The event *C* has P(*C*) = 0.20

The events *A* and *C* are mutually exclusive and the events *B* and *C* are statistically independent.

(c) Draw a Venn diagram to illustrate the events *A*, *B* and *C*, giving the probabilities for each region.

(5)

(d) Find P([*B* ∪ *C*]′)

(2)

(Total for question = 10 marks)

Q6.

A company has 1825 employees.
The employees are classified as professional, skilled or elementary.

The following table shows

- the number of employees in each classification
- the two areas, *A* or *B*, where the employees live

	A	B
Professional	740	380
Skilled	275	90
Elementary	260	80

An employee is chosen at random.

Find the probability that this employee

(a) is skilled,

(1)

(b) lives in area B and is not a professional.

(1)

Some classifications of employees are more likely to work from home.

- 65% of professional employees in both area A and area B work from home
- 40% of skilled employees in both area A and area B work from home
- 5% of elementary employees in both area A and area B work from home
- Event F is that the employee is a professional
- Event H is that the employee works from home
- Event R is that the employee is from area A

(c) Using this information, complete the Venn diagram below.

(4)

(d) Find $P(R' \cap F)$

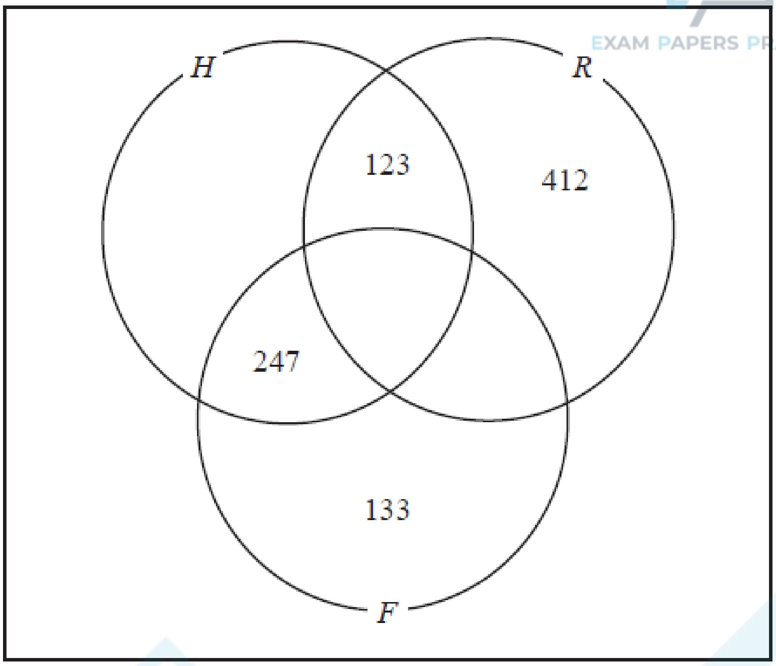
(1)

(e) Find $P([H \cup R]')$

(1)

(f) Find $P(F \mid H)$

(2)



(Total for question = 10 marks)